Hongyi Xu

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Research Interests

Computational and high-energy physics; statistical inference; Bayesian modeling; machine learning for particle and astrophysics research.

Education

University of Toronto **Sept 2023–June 2027 (expected)**
Major: Honours Bachelor of Science (HBSc), Physics Specialist Toronto, Canada

Selected completed coursework: Advanced Classical Mechanics; Advanced Physics Lab; Electricity & Magnetism; Thermal Physics; Electromagnetic Theory; Quantum Mechanics I; Introduction to Computer Programming; Introduction to Computer Science.

Selected coursework in progress: Computational Physics; Quantum Mechanics II; Nonlinear Physics; Introduction to High Energy Physics; Time Series Analysis; Relativistic Electrodynamics; Statistical Mechanics.

Research Experience

Neural-Network Event Selection and Statistical Analysis of Simulated Higgs Four-Lepton Events **May 2026–Present**
Research Project, advisor: Prof. Mario Campanelli, University College London Remote

- Studying neural-network event selection in public ATLAS simulated Higgs four-lepton signal and four-lepton background samples, focusing on the 2e2μ final state in an ongoing project.
- Constructed kinematic variables from lepton four-momenta and trained MLP classifiers with different input groups, using separate training, validation and evaluation subsets; examining selection effects on expected signal and background yields and four-lepton mass distributions.
- Developed profile-likelihood and event-group bootstrap workflows to investigate statistical uncertainty in simulation-based analyses.
- Code: https://github.com/hyi03/HiggsML

Research Experience

MaNGA Kinematics and Dark-Matter Halo Inference **2025–2026**

Independent Research

- Developed a rotation-curve mass-decomposition pipeline using Python and PyMC for 620 selected MaNGA disk galaxies, combining galaxy-level posteriors with population-level inference of the dark-matter halo concentration–mass relation.
- Applied MCMC and prior-corrected importance sampling to infer the relation's slope, normalization, and intrinsic scatter while characterizing halo mass–concentration degeneracy and parameter uncertainty.
- Tested robustness using prior predictive checks, prior-sensitivity analyses, sampling diagnostics, inclination-sensitivity tests, and PSIS diagnostics; recovered a modest negative concentration–mass slope ($\alpha = -0.093$) for the selected galaxies under fixed photometric inclinations, with subsample tests identifying inclination uncertainty as a major limitation.
- Manuscript: First-author manuscript (completed; not yet submitted): Bayesian Hierarchical Inference of the Dark Matter Halo Concentration–Mass Relation from MaNGA Disk Galaxy Rotation Curves (PDF); Project website: https://hyi03.github.io/manga-dm/

Technical Skills

- **Programming & Tools:** Python, PyMC, pyhf, LaTeX.
- **Statistical Methods:** Bayesian inference, MCMC, importance sampling, profile likelihood, bootstrap, posterior predictive checks, PSIS diagnostics.
- **Machine Learning:** MLP classification, feature-group comparison, model validation.
- **Languages:** Mandarin Chinese (native); English (pursuing an undergraduate degree taught in English).